

036015

Finite Elements Method for Engineering Analysis

Homework Assignment 4 - Numerical Integration

Winter 2024

Instructions

- Electronic submission of **typed** sheets only (PDF format through moodle).
- Submission file must be named: "*name surname.pdf*". No I.D number or homework number is required in the file name.
- Solution must contain all the relevant derivations, explanations, graphs and conclusions.
- All codes used must be added to the submission files.



In this assignment you will use the FE method to build a code that can do numerical integration over an arbitrary 2D domain. Integrating over a complex shape becomes easier when they are already meshed. You will use the shape functions and the Jacobian matrices of the elements to evaluate the integrand and the area of each element, and sum up the contributions from each element to get the approximate value of the integral.

The function that needs to be programmed:

```
int=integrateGeom(u,N,E)
```

u - is a vector containing all function values at each node, such that **u(i)** corresponds to the value of the function $u(x_i, y_i)$ where $N(i,:) = [x_i, y_i]$

N - $m \times 2$ matrix of the coordinates of each node in the global coordinate system.

E - $n \times 3$ matrix of the node numbers of each linear triangular element in the meshed shape.

Recommendation: When writing the code use descriptive and consistent variable names. Use external files to define the shape functions, and the quadrature rules, instead of hard-coding them in the main function.

1. Bonus [5pt]: Write a pseudocode or a flowchart that describes the algorithm of **integrateGeom** function.

For example: Loop over all elements in the geometry; Extract the node coordinates of the current element; Compute the Jacobian matrix and its determinant of the current element; Use the Gauss quadrature to compute the integral in the reference configuration... etc.

2. Write **integrateGeom** (in Matlab). [80pt]

3. Compute the area of a circle by using **integrateGeom** over the mesh from the previous assignment where **u=ones(length(N),1)**. [10pt]

This is equivalent to:

$$A = \iint_{\Omega} d\Omega$$

4. Analyze the error and the convergence of the numerical integration methods. Does the value converge to the analytical solution?

- (a) Plot a graph of the relative error for each level of refinement on a log-log scale. [5pt]
- (b) Will using more integration points in the Gauss integration algorithm provide a better convergence in this case? Why? [5pt]